24 Kiaria is 7 years older than Jay.
Martha is twice as old as Kiaria.
The sum of their three ages is 77

Find the ratio of Jay's age to Kiaria's age to Martha's age.

Jay: x

Kiaria: $x + 7$

Martha: $2(x + 7) = 2x + 14$

×	x	$+7$
2	$2x$	$+14$

1 mark
Express the three ages

$$x + (x + 7) + (2x + 14) = 77$$

1 mark
The three ages add to 77

$$4x + 21 = 77$$
$$-21 \qquad -21$$
$$4x = 56$$
$$\div 4 \qquad \div 4$$
$$x = 14$$

1 mark
Isolate x

Ages: 14, 21, 42

14 : 21 : 42

1 mark
Give the ratio in the requested order

(Total for Question 24 is 4 marks)